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# Effects of human-AI collaborative writing strategy on EFL students' argumentative writing skills

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## ABSTRACT

Human-AI collaborative writing has emerged as a new strategy that emphasizes the combined use of AI-assisted tools and human intelligence, integrating technological capabilities with critical thinking in the writing process. However, this strategy has not been empirically proven to be effective in enhancing students' argumentative writing skills, especially in the Indonesian EFL context. Thus, this study examined the effects of the human-AI collaborative writing strategy on Indonesian EFL students' argumentative writing skills and explored their perceptions about the strategy. This study used a pre-experimental one-group pretest-posttest. Thirty university freshmen were involved as participants. Data on students' writing skills were collected through an essay test, administered as both a pretest and posttest, while data on perceptions and experiences were gathered via a questionnaire. The written essays were assessed using rubrics adapted from Hamp-Lyons' Multiple-Trait (MT) and Facione's Holistic Critical Thinking Scoring Rubric (HCTSR), with the results subjected to statistical analysis via a paired sample *t*-test. The questionnaire responses were analyzed using descriptive statistics, with some quartile categorizations ranging from very negative to very positive. As a result, the effectiveness of the study could be seen from the improvement of the students' abilities to construct well-structured argument traits ( $t=54.87$ ,  $df=29$ ,  $p=0.000$ ,  $\eta^2=0.99$ ) and to use critical thinking skills ( $t=29.65$ ,  $df=29$ ,  $p=0.000$ ,  $\eta^2=0.96$ ) from the pre- and post-test scores. The implementation of the strategy fostered positive student perceptions and experiences, particularly regarding the ease of the writing process, writing enjoyment, and writing satisfaction, with mean approval ratings exceeding 85%. Hence, this study implies that the human-AI collaborative writing strategy can be used to train students on how to construct well-structured argument traits and convey critical thinking skills through written discourse using an intelligent technology-enhanced writing instruction.

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## KEYWORDS

Argument traits; argumentative writing skills; critical thinking skills; human-AI collaboration; writing strategy

## Introduction

Amidst the era of artificial intelligence (AI), Korteling et al. (2021) and Hernández (2019) underscore that there is a need to give human intelligence a space to contribute to completing course tasks, especially in writing argumentative essays. To address the need, Dhillon et al. (2024) and Luther et al. (2024) introduce a human-AI collaborative writing strategy, which has recently become the current research trend in Intelligent Computer-Assisted Language Learning (ICALL), especially in developing argumentative writing in the English as a Foreign Language (EFL) context. This strategy integrates AI-assisted tools (e.g. ChatGPT, Perplexity, Gemini, and QuillBot) to scaffold argumentative essay writing, while maintaining the essential role of human critical thinking in analyzing and evaluating the AI-generated content (Darwin et al., 2024; Newvine, 2023; Nguyen et al., 2024). For instance, students might use ChatGPT to generate a preliminary argumentative essay draft, then critically evaluate the AI's reasoning by identifying unsupported claims, adding scholarly evidence, and refining the draft, ensuring both technological scaffolding and human analytical rigor. These sequential analytical processes enable a human-AI collaborative writing strategy to cultivate critical thinking skills in argumentative writing contexts. Thus, this strategy integrates human cognition and artificial intelligence within ICALL-EFL environments through scaffolded argumentative writing tasks.

Besides the emphasis on a scaffolding approach (Dhillon et al., 2024; Zhang et al., 2023), the human-AI collaborative writing also promotes cognitive and process approaches (Yang et al., 2024) in EFL writing. Yang et al. (2024) state that human-AI collaborative writing equips students with rich inputs, which further require students' thinking skills to decide content for their desired arguments. This opportunity assists those with cognitive load problems to be able to developing the ideas for their writing. With various AI-generated content packed in the human-AI collaboration, students are encouraged to be cognitively versatile thinkers, which enable them to engage in a wide range of cognitive activities while preserving their own ideas (Dhillon et al., 2024; Yang et al., 2024). To attain the aim of enriching students' cognitive loads, a process needs to be emphasized, i.e. conducting cyclical writing phases to produce quality argumentative writing (Nguyen et al., 2024). The process may involve brainstorming writing topics, developing writing ideas, obtaining more supporting facts, and checking writing similarity (Fang et al., 2023; Marzuki et al., 2023; Yeh et al., 2024). Thus, the human-AI collaborative writing strategy facilitates diverse L2 writing approaches, demonstrating the symbiotic potential of technological and human cognitive integration.

However, several information inadequacies exist regarding the implementation of human-AI collaborative writing in producing argumentative essays. First, limited studies portray argumentative writing procedures by integrating human intelligence and AI. For instance, Song and Song (2023) and Wulyani et al. (2024) only explain that students mostly used ChatGPT and Google Translate to write their argumentative essays without mentioning which writing steps they used the AI-assisted tools and their human natural thinking skills. In addition, many similar studies only describe the types of AI-assisted tools used by students to finish their argumentative essay writing (Dwina, 2024; Filighera et al., 2022; Guo & Li, 2024; Ibrahim, 2023). An amount of systematic literature review has also been conducted in pointing out the model of human-AI collaborative writing strategy, but has not yet specifically covered instructional procedures for argumentative writing (Fang et al., 2023; Ji et al., 2023; Khakurel & Blomqvist, 2022). These facts consequently expect further studies to enclose the process of human-AI collaborative writing, especially in composing argumentative essays.

Second, to the best of the researchers' knowledge, very limited studies examined the effects of human-AI collaborative writing on EFL students' argumentative writing skills. For instance, Yang et al. (2025) found that although human-AI collaborative writing shows an effect on argumentative writing quality, this study's failure to delineate the sequential and frequential nature of the human and AI engagement leaves critical questions unanswered about the proper implementation of scaffolding and process approaches. Another study conducted by Yan (2024) also examined the effectiveness of human-AI collaborative writing in an L2 context. While the findings confirm a significant positive impact of human-AI collaborative writing on students' writing quality, this study's exclusive focus on processing ChatGPT-generated feedback for argumentative essay improvement presents notable limitations. A key limitation of this study lies in its insufficient details of both students' writing processes and teachers' scaffolding strategies during writing activities. Besides these two relevant studies, there are a number of studies dealing with AI-assisted argumentative writing (Apriani et al., 2024; Liu et al., 2024; Marzuki et al., 2023), however, they are extraneous to the present study's human-AI collaborative writing and critical thinking skills.

Third, some studies only depict students' perceptions and experiences for the use of AI-assisted tools in writing argumentative essays (Putra, 2023; Syahnaz & Fithriani, 2023; Wang, 2024) without exploring their perceptions and experiences on the use of human-AI collaborative writing. While the prior studies have shown students' positive perceptions and experiences with the use of AI-assisted tools, they have not examined the experiences when students were exposed to the writing

collaboration to make argumentative essays assisted by AI tools. Also, they have not provided the opportunity to experience the exposures of critical thinking practice during the writing activities. The availability of data on students' perceptions and experiences with human-AI collaborative writing indeed carries significant pedagogical implications for argumentative writing instruction. By exploring this empirical evidence, we know students' level of acceptance toward the strategy, particularly regarding writing enjoyment, self-efficacy, satisfaction, and process fluency, as well as their choice to recommend the use of the strategy to drill critical thinking skills. Such evaluative data directly inform instructors' decisions about whether to adapt, maintain, or discontinue the human-AI collaborative writing strategy in their teaching plans. Therefore, examining students' perceptions and experiences in human-AI collaborative writing is also crucial for developing an empirical basis to guide teachers' evidence-based implementation of this strategy in argumentative writing instruction.

Since these three research voids have not been addressed by prior studies, therefore, this study is conducted to analyze how human-AI collaborative writing strategy influences students' argumentative writing skills and explore their perceptions and experiences toward the strategy implementation. This study additionally demonstrates how the human-AI collaborative writing strategy is undertaken step-by-step in the method section as part of the treatment procedures. This study addresses the identified research gaps through the following research questions.

**Research question 1:** To what extent does the implementation of a human-AI collaborative writing strategy influence the argumentative writing skills of Indonesian EFL students?

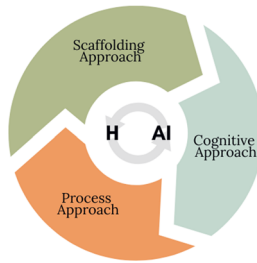
**Research question 2:** What are the perceptions and experiences of Indonesian EFL students regarding the use of a human-AI collaborative writing strategy in composing argumentative essays?

## Literature reviews

### ***Human-AI collaborative writing strategy: promoting an integrated-writing approach and critical thinking practices***

Human-AI collaborative writing strategy, a synergistic process where students and AI-assisted tools co-create text, is supported by an integrated-writing approach with technology-enhanced writing. [Figure 1](#) illustrates the conceptual framework of human-AI collaborative writing with an integrated-writing approach proposed in this study.

At a glance, this study uses the term *integrated-writing approach* to refer to an approach that accommodates other approaches used in writing instruction. This definition is proposed by Hyland (2003) to describe



**Figure 1.** Framework of human-AI collaborative writing with integrated-writing approach.

the most frequently used approach in L2 writing instruction. The following descriptions explain how human-AI collaborative writing accentuates the integrated-writing approach, especially in process, cognitive, and scaffolding approaches.

Regarding [Figure 1](#), the process approach aims to structure human-AI collaboration within a recursive writing cycle (Nguyen et al., 2024), emphasizing iterative refinement over linear progression. Students engage in repeated stages—planning, drafting, revising, and editing—to cultivate writing as a reflective, evolving practice (Nguyen et al., 2024; Zhang et al., 2023). Integrating AI tools (e.g. ChatGPT, QuillBot) at each stage enables real-time feedback, alternative phrasing, and content reorganization, fostering dynamic collaboration (Wang, 2024). For example, AI aids brainstorming during planning and identifies inconsistencies during revision, reinforcing writing as a recursive process (Susha et al., 2024). Through iterative drafting and AI-mediated feedback, students develop metacognitive skills by critically evaluating AI suggestions and making deliberate revisions (Nguyen et al., 2024; Zhang et al., 2023). This active engagement strengthens their authorial voice and critical thinking, ensuring AI complements rather than replaces cognitive effort. Thus, the process approach in human-AI collaborative writing not only enhances writing quality but also cultivates a mindset where writing is seen as an iterative, reflective, and adaptive process.

Human-AI collaborative writing strategy not only supports the process approach but also alleviates cognitive loads, facilitating a stronger emphasis on students' cognitive domain. Hyland (2003) states that the cognitive approach typically lies in the process of enriching students' knowledge so that they can produce a composition with relevant and tangible content. Human-AI collaborative writing strategy supports this approach in three key ways: reducing extraneous cognitive loads, enhancing metacognition through feedback loops, and scaffolding a complex writing process (emphasizing a scaffolding approach) (Yang et al., 2024). For example, L2 students can benefit from AI-assisted tools to help them when struggling with language use (e.g. grammar, vocabulary, and tenses) and content organization (e.g. coherence and argumentation). After the AI tools provide suggestions,

students can compare versions and justify choices, deepening their understanding of writing conventions (Yang et al., 2024). Moreover, their abilities to analyze, synthesize, interpret, and evaluate data generated by AI-assisted tools are drilled, placing an emphasis on critical thinking practices (Luther et al., 2024; Yang et al., 2024). This aligns with Vygotsky's (1978) concept of the Zone of Proximal Development (ZPD), where AI tools act as mediators, enabling learners to achieve higher competence by bridging gaps in linguistic and rhetorical knowledge.

At last, some studies believe that human-AI collaborative writing strategy promotes a scaffolding approach (Dhillon et al., 2024; Guo et al., 2024; Zhang et al., 2023). There are three practical scaffolding activities emphasized in the strategy, especially promoted in the present study: sentence-level scaffolding, structural-level scaffolding, and metacognitive scaffolding. In a sentence-level scaffolding, students are assisted in making a good sentence structure (Zhang et al., 2023). For instance, when they prompt 'you not okay', ChatGPT or Quillbot suggest 'you *are* not okay'. The omission of *are* in the students' prompt can be a new knowledge or feedback that can be used to scaffold a similar sentence in the future. In a structural-level scaffolding, students are assisted by AI tools to develop, for instance, an outline of a text (Lee et al., 2022; Zhang et al., 2023). In this case, students will be given an AI-generated outline, and they can choose which outline that suits their expectations. Of course, the outline is still editable, promoting students' decision-making aspect. In a metacognitive-level scaffolding, students are assisted by AI-assisted tools to critically rethink the developed essays, for instance. They may ask AI-assisted tools for feedback, but they can also have a dialogue about how the feedback should be addressed (Guo et al., 2024; Zhang et al., 2023). After the intense dialogue, students can revise independently, which emphasizes critical self-editing skills.

Despite the importance of the integrated-writing approach in enhancing L2 writing instruction quality, prior studies on human-AI collaborative writing strategy have notably neglected to incorporate this critical approach. Some studies only use double approaches, and a few others use a single approach. For instance, Dhillon et al. (2024) demonstrated that human-AI collaborative writing's scaffolding and cognitive approach simultaneously expands writing capacity and alleviates cognitive load, optimizing student focus on writing processes. While this study effectively addresses scaffolding and cognitive approaches, it lacks explicit delineation of how writing process activities are sequentially structured and implemented. Similarly, Guo et al. (2024) use scaffolding and process approaches but lack the emphasis on the cognitive approach. In this study, the researchers only focus on making a writing product, thus, they neglect the existence of cognitive drilling during the process. Moreover,



they also have no sequential explanations of writing stages that reflect the process approach. Similar to Guo et al. (2024), Zhang et al. (2023) only emphasize scaffolding and process approaches, making the undertaken strategy lack emphasis on the cognitive aspect. Even though Zhang et al. (2023) engage with the importance of reflective thinking skills, they have not explained how the process and scaffolding techniques they provide can accommodate the development of critical thinking skills. At last, several studies only use a single approach, which makes them lack complex discussion on the impacts of human-AI collaborative writing strategy (Li & Pan, 2024; Luther et al., 2024; Susha et al., 2024; Yan, 2024; Yang et al., 2024) on students' writing skills. The single-use approach studies have limited information due to incomplete discussion of the writing process involved.

By addressing the previous studies' theoretical gaps, especially on the use of writing approaches, this study promotes the use of an integrated writing approach to conduct a human-AI collaborative writing strategy to promote critical thinking practices. The use of this integrated approach also becomes the novelty of the present study because no previous study has used it in the context of human-AI-based writing strategy.

### ***Argumentative writing skills***

Argumentative writing skills refer to the ability to construct well-structured argument traits and use critical thinking skills (Guo et al., 2024; Pratama, 2015, 2018). In connection with the ability to construct well-structured argument traits, Pratama (2018) and Daniswara and Cahyono (2023) state that four essential argument traits must be used by students to make a good argument, namely, claim/assertion, reason, evidence, and inference/link back, which are integrated in the AREL argumentation technique. Guo et al. (2024) explain that a claim refers to an argument's assertion representing the arguer's position. A good claim should be supported by relevant data or reasons and warrants or evidence to strengthen the proposition of the argument (Daniswara & Cahyono, 2023; Guo et al., 2024). At last, an inference is provided to conclude the argument for each claim (Pratama, 2018). The structure of claim, reason, evidence, and inference cannot be shuffled to keep the logic of the argument (Guo et al., 2024). Regarding the ability to use critical thinking skills in writing argumentative essays, Crudele and Raffaghelli (2023) explain that critical thinking skills are required to ensure that the connection between the claim and the other argument traits does not carry logical fallacies. El Khoiri and Widiati (2017) also state that logical fallacies can hamper the logic of the argument and further weaken the proposition. Additionally, critical thinking skills are also used to process the information resources or references



before putting them as reason or evidence of argument traits. By appropriating the ability to construct well-structured argument traits and use critical thinking skills, students can produce more English-rhetoric argumentative writing than the oriental circular (Pratama, 2018), becoming the present study's focus to achieve.

Currently, perceiving argumentative writing skills to produce outstanding argumentative essays seems to be more challenging. It is due to the emergence of AI-assisted tools that provide an instant process for composing argumentative essays, such as ChatGPT, Gemini AI, and Perplexity (Fitria, 2024). The tools might make an argumentative essay along with the references according to the prompts used to construct the essay (Fitria, 2024). When coming to learning activities, students tend not to use their abilities in constructing a well-structured argument and using their critical thinking skills due to the emergence of instant argumentative writing (Fitria, 2024). Such situations should be seen as challenges in elevating students' argumentative writing skills, especially their critical thinking skills. Danry (2023) states that when students are not exposed to practicing critical thinking skills, they will have difficulties in solving problems, analyzing information, being independent, and initiating creativity. Thus, there should be a collaborative learning strategy that promotes the use of AI-assisted tools for writing argumentative essays and, at the same time, still fosters human critical thinking skills.

### ***Indonesian EFL context: using AI-assisted tools in argumentative writing***

Previous studies have documented the increasing integration of AI-assisted tools in English language learning contexts in Indonesia. For example, Wulyani et al. (2024) highlight the widespread adoption of platforms such as ChatGPT, Perplexity, Gemini, and DeepL Translate among Indonesian university students, particularly for generating ideas, brainstorming, developing content, conducting data analysis, drafting, correcting language accuracy, and managing literature. Similarly, other studies (Ananda et al., 2024; Dewi et al., 2024; Drajati et al., 2023) only demonstrate that the use of AI-assisted tools helps Indonesian EFL students complete their argumentative writing tasks by supporting them with cognitive loads. However, while these studies advocate for the utility of AI in supporting the writing process, they fail to delineate a structured instructional framework that outlines the sequential stages of an AI-assisted writing strategy. The absence of the framework information makes EFL teachers difficult how they design and conduct AI-based writing lesson plans. Moreover, existing literature does not sufficiently address how critical thinking skills are systematically cultivated within each phase of such a strategy. These gaps indicate a lack of emphasis on

scaffolding techniques in argumentative essay writing and the explicit integration of critical thinking practice. Consequently, the pedagogical application of AI-assisted tools in Indonesian EFL writing instruction remains underdeveloped, lacking a well-defined instructional design.

Despite growing global research on human-AI collaborative writing, a significant gap remains in understanding its application within Indonesia's English as a Foreign Language (EFL) context. To the best of the researchers' knowledge, no existing study has investigated how Indonesian learners and educators integrate AI tools like ChatGPT or Quillbot into human-AI collaborative argumentative writing strategy. The absence of the localized research risks promoting AI integration models that overlook key factors, such as: (1) the prevalence of teacher-centered instruction in Indonesian schools, which may require tailored professional development for collaborative AI pedagogy; and, the need for ethical guidelines addressing generative AI use in a context where plagiarism perceptions differ culturally. Filling this gap would provide empirically grounded strategies for balancing technological benefits with the preservation of critical thinking skills in Indonesia's EFL context.

Referring to the above literature reviews, the present study believes that examining the impacts of human-AI collaborative writing strategies on Indonesian EFL students' argumentative writing skills is justifiable. By referring to the previous studies, this study defines argumentative writing skills as Indonesian EFL students' ability to construct well-structured argument traits and use critical thinking. As part of supporting this pre-experimental study as well as addressing the inadequate information on how human-AI collaborative writing strategy is mostly undertaken, the teaching steps of the strategy are also provided in this study's methods. In addition, this study reveals Indonesian EFL students' perceptions after using a human-AI collaborative writing strategy to look at their perceived learning experience. This second objective is necessary since many studies argue that examining students' responses toward the undertaken learning strategy is important to support enlightening the effectiveness of the strategy, adapting learning, increasing student engagement, encouraging self-reflection, and creating a democratic learning setting (Alawadhi & Abu-Ayyash, 2021; Rahmania & Mandasari, 2021).

Finally, the findings of this study offer valuable insights for Indonesian EFL teachers seeking to integrate AI into their pedagogical practices. First, the study provides a framework for implementing a human-AI collaborative writing strategy, which is essential for fostering an AI-enhanced learning environment in contemporary education. Teachers will gain practical guidance on structuring writing activities that not only leverage AI tools but also cultivate students' critical thinking skills. Specifically, students can engage in brainstorming, evaluating source reliability, and data mining to

identify pertinent evidence and examples for argumentative essays. Second, the study delineates the respective roles of teachers and students in each phase of the learning process, ensuring clarity in instructional expectations. This structured approach facilitates dynamic writing instruction by incorporating scaffolding techniques, thereby supporting learners in progressively developing their argumentation skills. Third, the study equips teachers with strategies for assessing critical thinking as demonstrated in student writing, enabling them to evaluate both content and cognitive engagement effectively. Consequently, this study holds significant practical implications, empowering EFL teachers to design, implement, and refine argumentative writing instruction through a human-AI collaborative writing strategy while systematically fostering critical thinking.

## Methods

This study used a pre-experimental one-group pretest-posttest to examine the effects of the human-AI collaborative writing strategy on Indonesian EFL students' argumentative writing skills and to explore the students' perceptions and experiences of writing argumentative essays using the human-AI collaborative writing strategy. There were thirty participants registered as active members of a debate community who came from various study programs at one of the public universities in Surabaya, East Java Province, Indonesia. The debate community was an extracurricular activity aimed at developing critical thinking skills in both spoken and written forms using the English language. The activities were conducted weekly with a minimum of two meetings per week. The participants' English proficiency scores ranged from 450 to 500, corresponding to low-intermediate to high-intermediate levels. The proficiency test was administered using a standardized test developed by the university's language unit. The test specifications and blueprint were designed to align closely with the Test of English as a Foreign Language (TOEFL), as established by Educational Testing Service (ETS). This test was a mandatory requirement for incoming freshmen prior to their enrollment in programming courses during the first semester of study. Additionally, they self-reported that they were technologically literate; thus, they knew how to use a laptop and a handphone/tablet, had been familiar with several generative AI tools (e.g. ChatGPT and Google Translate), understood how to connect to the internet, and had been accustomed to brows digital materials through different web browsers (e.g. Mozilla Firefox, Safari, Opera, and Google Chrome). More importantly, since they were passionate about debate, they understood that debaters always wrote argumentative essays to conduct a debate. The writing activities were

highly intensive, as participants were required to compose argumentative essays in Microsoft Word format prior to delivering debate speeches. Each meeting included two debate practice sessions, and during each session, students completed one argumentative essay. Prior to the present study, participants had limited training in utilizing AI-assisted tools for argumentative essay composition. Table 1 presents the demographic characteristics of the participants.

This study used three research instruments to support data collection procedures, namely an essay test with scoring rubrics adapted from the Hamp-Lyons' Multiple-Trait (MT), Facione's Holistic Critical Thinking Scoring Rubric (HCTSR), and a response questionnaire. The essay test was administered as a pretest and post-test, consisting of similar instructions, namely composing an argumentative essay (whether pros or cons) about the use of AI in completing school tasks (900 to 1200 words). The researchers' rationales for taking the same essay topics were to ensure cognitive/content comparability, minimize topic-related variability, and accurately measure writing skill development. The familiarity of the contents might occur, but this was not the focus of the study; thus, future research may consider this aspect for forthcoming research variables. The essay test should be completed in thirty minutes, which had become the internationally standardized case-building time to compose argumentative essays in any Asian and Australian Parliamentary debates. The test instruction had been validated and examined for its readability by related experts and random selected students through interviews. Then, the test was scored based on a developed scoring rubric adapted from the Hamp-Lyons' Multiple-Trait (MT) and Facione's Holistic Critical Thinking

**Table 1.** Demography Data of Participants.

| Aspects                                                 | <i>N</i> | <i>f</i> |
|---------------------------------------------------------|----------|----------|
| Gender                                                  |          |          |
| • Male                                                  | 19       | 63.33%   |
| • Female                                                | 11       | 36.67%   |
| Age                                                     |          |          |
| • 18 years old                                          | 1        | 3.33%    |
| • 19 years old                                          | 24       | 80%      |
| • 20 years old                                          | 4        | 13.34%   |
| • 21 years old                                          | 1        | 3.33%    |
| Semester                                                |          |          |
| • First semester                                        | 30       | 100%     |
| Study programs                                          |          |          |
| • English education                                     | 12       | 40%      |
| • English literature                                    | 4        | 13.34%   |
| • Law                                                   | 5        | 16.66%   |
| • Science and Mathematics                               | 9        | 30%      |
| Debate experience (from the first-time learning debate) |          |          |
| • < 3 years                                             | 29       | 96.67%   |
| • > 3 years                                             | 1        | 3.33%    |

Scoring Rubric (HCTSR) (CVR = +1.00, CVI = 1.00). At last, the response questionnaire was used to reveal the participants' writing experiences on the use of the Human-AI collaborative writing strategy (Cronbach's  $\alpha$  = .89). The questionnaire used a six-point Likert scale from strongly disagree (1) to strongly agree (6). Table 2 shows the blueprint of the rubric items used in MT and HCTSR, and Table 3 describes the blueprint of the response questionnaire.

The students' essays obtained both in the pretest and post-test were scored by three raters. This study employed both intra- and inter-rater reliability to examine the consistency of the scores given in the pretest and post-test. In addition, the obtained data were analyzed using descriptive statistics and paired sample *t*-test. Descriptive statistics was used to describe students' argumentative writing skills before and after implementing the human-AI collaborative writing strategy and portray their response toward the strategy implemented. The results of the MT and HCTSR descriptive statistics were explained according to Hamp-Lyons and Henning's (1991) and Facione's (1994) theories, respectively (see Appendix A1). Meanwhile, a paired sample *t*-test was used to examine the effectiveness of human-AI collaborative writing. To interpret the effect size, this study referred to the conversion of the Eta squared value ( $\eta^2$ ) proposed by Cohen (1988) and Pallant (2020) from small (.01) to large effect (0.14). In addition, the data of students' responses were analyzed using a descriptive statistic and were categorized using adapted Pratama's quartile categorization (2024): Q1=very negative ( $M\% < 25\%$ ), Q2=negative ( $25\% \leq M\% < 50\%$ ), Q3=positive ( $50\% \leq M\% \leq 75\%$ ), and Q4=very positive ( $M\% > 75\%$ ). This study used IBM SPSS 25.0 to do the statistical tests. This study employed methodological triangulation by incorporating qualitative analysis of students' essays to corroborate the quantitative findings. The qualitative analysis followed the framework established by Pratama (2018), which involved systematic identification and evaluation of four key argument components or traits: Claim (C), Reason (R), Example (E), and Inference/Link back (I). For coding purposes, sequential numbering was applied to recurring components (e.g. C1 for the first claim, C2 for the second claim). The analytical process comprised two main phases: First, we examined the logical structure of arguments by assessing both the sequential arrangement of these components (C-R-E-I) and their conceptual coherence. Second, we evaluated the overall argument quality based on the demonstrated critical thinking skills, specifically analyzing the strength of logical connections between components and the robustness of supporting evidence. This dual-phase analysis enabled comprehensive judgments regarding the structural integrity and critical thinking evident in the argumentative essays and validated the present study's results.

**Table 2.** Blueprint of Scoring Rubric (MT and HCTSR).

| Aspects | Objectives                                                     | Sub-skills                                                                          | Scores | References                                                               |
|---------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------|
| MT      | Assessing the structure of argument-trait quality              | 1. Communicative quality                                                            | 1–9    | Hamp-Lyons & Henning (1991); Daniswara and Cahyono (2023); Pratama, 2018 |
|         |                                                                | 2. Interestingness                                                                  | 1–9    |                                                                          |
|         |                                                                | 3. Referencing                                                                      | 1–9    |                                                                          |
|         |                                                                | 4. Organization (the structure of claim, reason, evidence, and inference)           | 1–9    |                                                                          |
|         |                                                                | 5. Linguistic accuracy (grammar, vocabulary, spelling, and punctuation)             | 1–9    |                                                                          |
| HCTSR   | Assessing the critical thinking skills reflected in the essays | 1. Accurately supporting the claim with relevant reasons, evidences, and inferences | 1–4    | Facione (1994); Yin et al. (2023)                                        |
|         |                                                                | 2. Avoiding logical fallacies                                                       | 1–4    |                                                                          |
|         |                                                                | 3. Providing and explaining alternatives and their possible impacts                 | 1–4    |                                                                          |
|         |                                                                | 4. The coherence of the logic between paragraphs                                    | 1–4    |                                                                          |
|         |                                                                | 5. Drawing non-fallacious conclusions                                               | 1–4    |                                                                          |

**Table 3.** Blueprint of Response Questionnaire.

| Aspects                 | Sub-aspects                                                                                       | Item Number | References                               |
|-------------------------|---------------------------------------------------------------------------------------------------|-------------|------------------------------------------|
| Writing experience      | Use of AI-assisted tools, exposure of human-AI collaborative writing, easiness, and effectiveness | 1–5         | Bulqiyah et al. (2021), Li et al. (2020) |
| Emotions during writing | Enjoyment, efficacy, and satisfaction                                                             | 6–8         |                                          |
| Recommendation          | Recommending the use of human-AI collaborative writing                                            | 9–10        |                                          |

### ***Teaching and learning process using human-AI collaborative writing strategy***

This study employed a three-meeting pre-experimental design conducted over one week. The researchers acted as the teachers in this study scenario. During the first meeting, participants composed argumentative essays without AI-assisted tools (e.g. ChatGPT) to establish a baseline of their independent writing capabilities. To ensure protocol adherence, researchers monitored this 30-minute writing session, which was conducted using pencil and paper. This manual writing condition served two purposes: (1) to assess students' unaided argumentative writing skills, and (2) to provide comparative data for subsequent AI-assisted sessions. No

interventions were implemented during this session, allowing participants to rely solely on their prior knowledge and experience. Following completion, these baseline essays were collected and designated as pre-test materials. Participants were then briefed about the subsequent two sessions, which would involve a human-AI collaborative writing strategy. We provided an overview of the strategy's theoretical rationale, practical implementation, and potential future applications in academic contexts, particularly for competitive debate and argumentative writing scenarios. The subsequent sessions employed laptops with internet access and AI tools, enabling direct comparison between unaided and AI-assisted writing processes to examine participants' perceptual and experiential differences.

In the second meeting, we started implementing the human-AI collaborative writing strategy. The strategy consisted of four learning phases comprising pre-, while, post-, and outside-learning activities with ten key steps of the teaching and learning process. The strategy was implemented by adopting to the integrated writing approach. Figure 2 shows the mapping of the instructional phases during the treatment period.

This study implemented a human-AI collaborative writing strategy for two meetings, with 120 min allotted for each meeting. In the pre-learning activities, participants learned the types and functions of ChatGPT, Gemini, Google Translate, DeepL, Quillbot, and Plagiarism Checker, along with how to use them. They also learned how to make a good argumentative essay using appropriate critical thinking skills and knew its writing purpose (emphasizing genre-based approach). After that, they experienced a mock-up session using a practice topic for writing argumentative essays using the strategy. This topic was chosen by the participants due to their familiarity and interest (topic: *Should we prohibit smoking in public places?*). In this case, the researchers' role was a facilitator of the brief training-like session. We guided how to make a good prompt for ChatGPT and Gemini, how to search for relevant content using ChatGPT and Gemini, how to do translation using Google Translate and DeepL, how to paraphrase using Quillbot, how to revise and edit using ChatGPT and Gemini, and how to check similarity level using

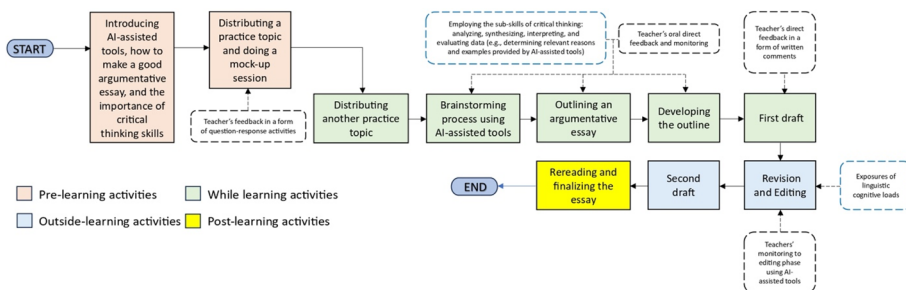


Figure 2. Instructional phases using a human-AI collaborative writing strategy.



Plagiarism Checker. Moreover, we should provide responses and feedback to ensure the initial understanding of the short training-like activities. This pre-learning session could last sixty minutes.

After ensuring the participants' complete understanding, they were given another practice topic (e.g. *No love at first sight*). They were given ten minutes to brainstorm ideas, five minutes to outline, and fifteen minutes to develop the outline; of course, they must use AI-assisted tools to help them find relevant content to make the first draft (emphasizing a cognitive approach). In addition, they could control the ideas they wanted to argue, so that this step also emphasized their thinking abilities. During these processes, participants also learned sub-skills of critical thinking, such as analysis (evaluating AI-generated content), synthesis (combining ideas), interpretation (understanding data), and evaluation (judging argument quality). These sub-skills could be conducted in two or more cycles, depending on how satisfied participants were with the arguments they constructed using the assistance of AI tools (emphasizing personalized learning). Additionally, they should consider several elements in writing argumentative essays, such as the title, the claims of arguments for each paragraph, reasons and evidence for each determined claim, inferences, and a list of resources. The numbers of the paragraphs were extraneous to the present study's concerns as long as the participants wrote 900–1,200 words in total. In these writing steps, the researcher's support was in the form of oral direct feedback and monitoring service to ensure the participants worked on their argumentative writing phases well. In other words, we should not interfere with the writing too much and let the participants engage with AI-assisted tools to construct their essays (emphasizing scaffolding and process approaches). The rest thirty minutes in the first meeting were used by the researchers to provide written feedback to enhance the quality of the linguistic accuracy (emphasizing cognitive and text-based approaches), organization, and critical thinking skills.

After the second meeting was done, the participants could continue to revise and edit the first draft that had been commented on by the researchers outside the classroom learning process. To support their autonomous writing process, the communication between participants and researchers was undertaken through group WhatsApp chats. At outside-learning activities, the participants revised their linguistic mistakes using Quillbot, ChatGPT, or Gemini. In this case, they could learn the correct form of language use (emphasizing a text-based approach). Another activity was to revise and edit the contents of their essays using ChatGPT or Gemini to deal with the researchers' feedback for their critical thinking skills reflected in their essays. After they had revised and edited the essays, they should check the degree of similarity using

Plagiarism Checker. This step was important to teach them to build their academic ethics when citing several resources. After completing these writing steps, they could reread and submit the final essay to the researchers. The submitted essays were used as their learning portfolios and were not used as the post-test essays.

In the third meeting, the participants began with a small discussion while we distributed a questionnaire assessing their perceptions and experience using the human-AI collaborative writing strategy. This activity was undertaken casually to make sure the participants reflected their thoughts and feelings in a comfortable setting. After completing the questionnaire, the participants were given a topic that was similar to the pre-test topic and asked to develop an argumentative essay. They were given thirty minutes to construct the essay using a human-AI collaborative writing strategy. This post-test essay was assessed using the Hamp-Lyons' Multiple-Trait (MT) and Facione's Holistic Critical Thinking Scoring Rubric (HCTSR). The obtained scores were used as the post-test scores, which were further compared with the pre-test scores.

### **Research ethics**

This study was conducted by considering important aspects of research ethics. This study employed a consent form distributed to the participants. Each completed form reflected the participants' willingness to contribute to the study without political oppression. The consent form was distributed before conducting the research, after socializing the objectives of the research and how the participants could be part of it. The detailed information of the participants was not disclosed to keep the essence of confidentiality. Every data obtained from the field was stored in the researchers' database so no parties could access it. Since this study used humans as research subjects, they had the right to withdraw from the study when they felt uncomfortable and oppressed. At last, this study used non-premium AI-assisted tools so every participant could access them. By signing up for accounts on some AI tool websites, the participants agreed to the terms and conditions provided by the providers.

### **Results**

#### ***Effects of human-AI collaborative writing strategy on students' argumentative writing skills***

This study used a descriptive statistic and paired sample t-test to address the first research question. The results of the tests were reported by, first, portraying the participants' argumentative writing skills before and after the treatment and, second, depicting the effects of the human-AI

collaborative writing strategy on their argumentative writing skills. Table 4 showcases the participants' argumentative writing skills before and after the undertaken treatment.

The  $M$  scores in Table 4 were obtained from the inter-raters with consistent and reliable intra-raters' scores analyzed using descriptive statistics. The  $M$  values of argumentative writing skills were aggregated from the  $M$  scores obtained from the Hamp-Lyons' Multiple-Trait (MT) and Facione's Holistic Critical Thinking Scoring Rubric (HCTSR), with a maximum score of 65. Table 4 explains that there was a different outcome after the treatment; students' argumentative writing skills moved from the weak ( $M=18.80$ ,  $SD=4.18$ ) to the good quartiles ( $M=56.87$ ,  $SD=4.18$ ) (see Appendix A1 for weak and good categorization). This finding implies that the raters agreed that a human-AI collaborative writing strategy contributed to improving students' abilities to construct well-structured arguments and to use critical thinking skills. However, Table 4 was only a superficial report from the raters' scores based on the pre-test and post-test scores. To deepen the analysis of the effect of the undertaken human-AI collaborative writing strategy, a paired sample  $t$ -test was able to be carried out due to no violation of the assumption test. Table 5 shows the result of the paired sample  $t$ -test based on the aggregate and sub-skill analysis.

Table 5 conveys that there was a significant increase in the students' argumentative writing skills through the application of the human-AI collaborative writing strategy ( $M=38.07$ ,  $SD=3.99$ ,  $t(29) = 52.13$ ,  $p < 0.05$ ). This study also found that each sub-skill in both the ability to construct well-structured arguments and conveying critical thinking skills also significantly increased. According to Cohen (1988) and Pallant (2020), since all Eta squared values ( $\eta^2$ ) were greater than 0.14, the study confirmed that there was a very big effect of the undertaken human-AI collaborative writing strategy in improving students' argumentative writing skills ( $\eta^2 > 0.50$ ). This means that the undertaken human-AI collaborative writing strategy could increase students' argumentative writing skills: a sub-skill to construct a well-structured argument and a sub-skill to convey critical thinking. As the triangulation data of the quantitative data above, Table 6 shows an example of argument contrast before and after the implementation of the human-AI collaborative writing strategy made by a student. The contrast was obtained by adapting Pratama's (2018) argument analysis.

In short, Table 6 shows that a student had successfully transformed the patterns of the argument traits from cyclical into more well-structured. This condition affected the student's ability to expand more content to support the given claim, as proven by the addition of word counts. We took this example as it represented the most commonly used pattern in

**Table 4.** Indonesian EFL Students’ Argumentative Writing Skills: Before and After Treatment.

| Aspects and Subskills                                                              | Before Treatment |      |                                                                                                                                                                                                          | After Treatment |      |                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------|------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                    | M                | SD   | Meaning                                                                                                                                                                                                  | M               | SD   | Meaning                                                                                                                                                                                                                                                                         |
| Hamp-Lyons’ Multiple-Trait (MT)                                                    | 12.47            | 2.93 | The argumentative writing skills showed no ability to communicate, no examples, no recognized organizational structure, and no control of linguistic accuracy, but occasionally showed interesting idea. | 39.10           | 0.96 | The argumentative writing skills showed an ability to communicate without causing comprehension difficulties; sufficient creativity, novelty, and interest; frequent suitable examples; good logical organizational structure; and no significant errors of linguistic accuracy |
| • Communicative quality                                                            | 1.60             | 0.62 |                                                                                                                                                                                                          | 7.70            | 0.47 |                                                                                                                                                                                                                                                                                 |
| • Interestingness                                                                  | 6.40             | 1.22 |                                                                                                                                                                                                          | 7.63            | 0.67 |                                                                                                                                                                                                                                                                                 |
| • Referencing                                                                      | 1.33             | 0.61 |                                                                                                                                                                                                          | 7.77            | 0.62 |                                                                                                                                                                                                                                                                                 |
| • Organization (the structure of claim, reason, evidence, and inference)           | 1.47             | 0.68 |                                                                                                                                                                                                          | 8.67            | 0.48 |                                                                                                                                                                                                                                                                                 |
| • Linguistic accuracy (grammar, vocabulary, spelling, and punctuation)             | 1.67             | 0.88 |                                                                                                                                                                                                          | 7.33            | 0.48 |                                                                                                                                                                                                                                                                                 |
| Facione’s Holistic Critical Thinking Scoring Rubric (HCTSR)                        | 6.33             | 1.75 | The argumentative writing skills showed significantly weak to unacceptable critical thinking skills.                                                                                                     | 17.76           | 1.07 | The argumentative writing skills showed acceptable to strong critical thinking skills.                                                                                                                                                                                          |
| • Accurately supporting the claim with relevant reasons, evidences, and inferences | 1.30             | 0.47 |                                                                                                                                                                                                          | 3.53            | 0.51 |                                                                                                                                                                                                                                                                                 |
| • Avoiding logical fallacies                                                       | 1.27             | 0.45 |                                                                                                                                                                                                          | 3.47            | .051 |                                                                                                                                                                                                                                                                                 |
| • Providing and explaining alternatives and their possible impacts                 | 1.23             | 0.43 |                                                                                                                                                                                                          | 3.00            | 0.64 |                                                                                                                                                                                                                                                                                 |
| • The coherence of the logic between paragraphs                                    | 1.13             | 0.35 |                                                                                                                                                                                                          | 3.77            | 0.43 |                                                                                                                                                                                                                                                                                 |
| • Drawing non-fallacious conclusions                                               | 1.40             | 0.50 |                                                                                                                                                                                                          | 4.00            | 0.00 |                                                                                                                                                                                                                                                                                 |
| Argumentative writing skills                                                       | 18.80            | 4.18 | Weak ability in constructing well-structured argument and critical thinking skills                                                                                                                       | 56.87           | 1.55 | Good ability in constructing well-structured argument and critical thinking skills                                                                                                                                                                                              |

**Table 5.** Results of paired sample *T*-test.

| Pair | Aspects (before and after treatment)                                             | <i>M</i> | <i>SD</i> | <i>t</i> | <i>df</i> | <i>p</i> | $\eta^2$ |
|------|----------------------------------------------------------------------------------|----------|-----------|----------|-----------|----------|----------|
| 1    | Argumentative writing skills (Aggregate score)                                   | 38.07    | 3.99      | 52.13    | 29        | 0.000    | 0.96     |
| 2    | Ability to construct well-structured arguments (Subskill 1)                      | 26.63    | 2.66      | 54.87    | 29        | 0.000    | 0.99     |
| 3    | Communicative quality                                                            | 6.10     | 0.84      | 39.55    | 29        | 0.000    | 0.98     |
| 4    | Interestingness                                                                  | 1.23     | 1.04      | 6.50     | 29        | 0.000    | 0.59     |
| 5    | Referencing                                                                      | 6.43     | 0.82      | 43.12    | 29        | 0.000    | 0.98     |
| 6    | Organization                                                                     | 7.20     | 0.81      | 48.98    | 29        | 0.000    | 0.98     |
| 7    | Linguistic accuracy                                                              | 5.67     | 0.99      | 31.29    | 29        | 0.000    | 0.97     |
| 8    | Ability to convey critical thinking skills (Subskill 2)                          | 11.43    | 2.11      | 29.65    | 29        | 0.000    | 0.96     |
| 9    | Accurately supporting the claim with relevant reasons, evidences, and inferences | 2.23     | 0.77      | 15.81    | 29        | 0.000    | 0.89     |
| 10   | Avoiding logical fallacies                                                       | 2.20     | 0.66      | 18.14    | 29        | 0.000    | 0.91     |
| 11   | Providing and explaining alternatives and their possible impacts                 | 1.77     | 0.73      | 13.30    | 29        | 0.000    | 0.85     |
| 12   | The coherence of the logic between paragraphs                                    | 2.63     | 0.57      | 25.94    | 29        | 0.000    | 0.95     |
| 13   | Drawing non-fallacious conclusions                                               | 2.60     | 0.50      | 28.58    | 29        | 0.000    | 0.96     |

improved argumentative essays. To provide more detailed information, Figure 3 recaps the most used argument trait patterns in improved essays, along with the trend of the word count expansion and the total number of paragraphs. Additionally, Figure 4 shows the themes of each argument trait used in the improved essays.

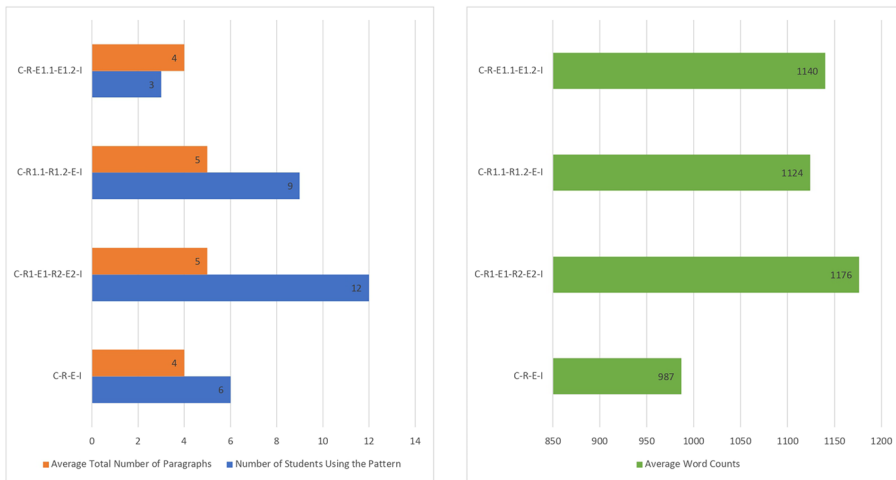
Figure 3 explains that the argument pattern mostly used C-R<sub>1</sub>-E<sub>1</sub>-R<sub>2</sub>-E<sub>2</sub>-I (*N*=12), which could develop the arguments around five paragraphs with 1176 words, on average. The data implied that the C-R<sub>1</sub>-E<sub>1</sub>-R<sub>2</sub>-E<sub>2</sub>-I became the best pattern of argument traits to produce more elaborated argument content with an emphasis on critical thinking skills. However, no student used the C-R<sub>1</sub>-E<sub>1</sub>-R<sub>2</sub>-E<sub>2</sub>-I pattern to convey ideas in support of the topic of the essay. In other words, all of them rejected the use of AI in completing school tasks. To look at how the themes of each argument trait were used in the improved essays, Figure 4 shows the trend of the argument flows conveyed by the students.

Based on Figure 4, the flow diagram outlines a structured approach to developing an argumentative essay with a negative stance on using AI tools in academic tasks. It began by establishing the central claim that these tools caused more harm than good in learning, supported by four other key claims: erosion of critical cognitive development, compromised academic integrity, equity disparities, pedagogical obsolescence, and long-term

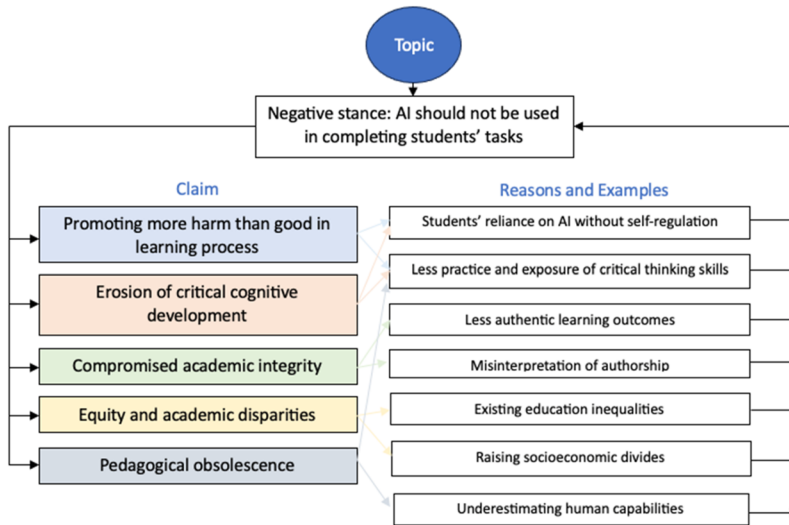
**Table 6.** Example of argument contrast before and after the human-AI collaborative writing strategy.

| argument (before)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | contrastive aspect                                        | argument (after)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Artificial intelligence bring more harm than good if it will implemented on the class ( $C_1$ ). The students cannot focus to the materials because they busy with operating artificial intelligence ( $R_1$ ). The teachers also difficult to monitor the students because they are so many ( $R_2$ ). Artificial intelligence also has many negative side ( $C_2$ ). AI is technology and technology can be affected by virus ( $C_3$ ). So, artificial intelligence cannot help students to learn for long periods ( $I$ ). (69 words) | <i>Example of one argument</i>                            | Artificial intelligence (AI) tools promote more harms than goods for the learning process ( $C$ ). Some students might be dependent to the use of AI tools ( $R_1$ ). For instance, they always consult the materials and learning tasks to ChatGPT without considering their natural thinking skills ( $E_1$ ). This dependency leads them to be instant thinkers rather than critical thinkers ( $R_2$ ). Such longer use of AI tools can hamper the progression of drilling critical thinking skills through learning process. Consequently, students cannot be independent in solving their problems occurred both inside or outside the classroom learning process ( $E_2$ ). Therefore, AI tools bring more harms to the students' learning process ( $I$ ). (100 words) |
| The argument did not display a good communication because the unstructured argument traits. There was no examples or evidences to strengthen the reasons and the claim. Instead of providing evidence, there were two new claims ( $C_2$ and $C_3$ ) that were extraneous to the first claim ( $C_1$ ). Thus, the inference ( $I$ ) is difficult to accept since it did not explicitly confirm the three claims. The messy organizational structure created not interesting idea. Moreover, there were many grammatical errors.           | <i>Analysis based on argument traits structure (MT)</i>   | The argument organizational structure was good since it covered claim, reason, evidence, and inference. The two reasons were supported by each evidence. The inference was made relevantly to strengthen the claim asserted in the beginning. The argument was also more interesting and easier to follow with good linguistic accuracy. With this good MT aspect, the argument even could be more elaborated compared to the before one.                                                                                                                                                                                                                                                                                                                                      |
| Since the MT aspect was not met, the logic did not exist within the argument. There were several logical fallacies caused by unclear premises. The reasons were less contributing to the claim so the argument was not strongly weak.                                                                                                                                                                                                                                                                                                     | <i>Analysis based on critical thinking skills (HCTSR)</i> | Since the MT aspect was met, the logic existed within the argument. The reasons and evidences were suitable and could strengthen the claims. The inference could sum up the explanations and strengthen the claim too. This made the argument easy to be followed without ambiguity and logical fallacies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

skill deficits. Each claim was reinforced by reasons and examples that provided evidence and contextualized the argument. For instance, the diagram highlighted how overreliance on tools led to *less authentic training outcomes* (e.g. students producing AI-generated work without mastering concepts) and exacerbated *existing education inequalities* (e.g. privileging students with access to advanced technology). Figure 4 also confirms the



**Figure 3.** Most occurring patterns of argument traits in the improved essays.



**Figure 4.** Flow diagram of themes used in the improved argumentative essays.

C-R<sub>1</sub>-E<sub>1</sub>-R<sub>2</sub>-E<sub>2</sub>-I pattern when developing argumentative essays conveyed by students. It was proven by a single claim that was followed by double reasons and evidence, then followed by a collective inference to link back to the stance. The flow diagram thus served as a blueprint for constructing a cohesive argument, emphasizing cause-and-effect relationships and empirical grounding to challenge the use of tools in an academic setting.

Therefore, by looking at the above quantitative and qualitative data, the collaboration of human intelligence and artificial intelligence could hold up the structure of the argument traits with relevant, rich, and critical content of the argument. This concluded that the effect of human-AI collaborative writing strategy on the students' argumentative skills was proven.



**Table 7.** Students' perceptions and experiences.

| Subskill                | Item Number | Aspects                                                             | <i>M</i> | <i>SD</i> | <i>M%</i> | Pratama's Quartile Category |
|-------------------------|-------------|---------------------------------------------------------------------|----------|-----------|-----------|-----------------------------|
| Writing experience      | 1           | Personalized learning using AI-assisted tools                       | 5.17     | 0.65      | 86.16     | Q4                          |
|                         | 2           | Exposure to critical thinking skills                                | 5.03     | 0.72      | 83.83     | Q4                          |
|                         | 3           | Exposure to human-AI collaborative writing                          | 5.87     | 0.34      | 97.83     | Q4                          |
|                         | 4           | Easiness in writing process                                         | 5.93     | 0.25      | 98.83     | Q4                          |
|                         | 5           | Effective writing process                                           | 5.90     | 0.30      | 98.33     | Q4                          |
| Emotions during writing | 6           | Writing enjoyment                                                   | 5.73     | 0.45      | 95.50     | Q4                          |
|                         | 7           | Writing efficacy                                                    | 5.50     | 0.51      | 91.66     | Q4                          |
|                         | 8           | Writing process satisfaction                                        | 5.70     | 0.47      | 95.00     | Q4                          |
| Recommendations         | 9           | Recommending the use of human-AI collaborative writing strategy     | 5.97     | 0.18      | 99.50     | Q4                          |
|                         | 10          | Recommending the involvement of AI in argumentative writing process | 6.00     | 0.00      | 100       | Q4                          |

### ***Students' responses to the strategy implementation***

To address the second research question, this study conducted a descriptive statistic to reveal the students' responses after using the human-AI collaborative writing strategy. Table 7 shows the students' writing experience, emotions during writing, and recommendations after implementing the proposed strategy.

Table 7 describes that the students showed a very positive response to using the human-AI collaborative writing strategy to help them compose argumentative essays. All of them recommended the involvement of AI-assisted tools in writing argumentative essays since they offered writing easiness and effectiveness. Moreover, they also agreed that the involvement of AI-assisted tools in argumentative writing promoted personalized learning, writing enjoyment, efficacy, and satisfaction. They also showed a very positive response to the use of the human-AI collaborative writing strategy to give them exposure to critical thinking skills.

### ***Discussion***

This study found that the human-AI collaborative writing strategy effectively enhanced students' argumentative writing skills, encompassing the ability to construct well-structured argument traits and convey critical thinking. First, this study showed that there was a significant improvement

in the students' ability to construct well-structured arguments. The human-AI collaborative writing strategy allowed the students to show quality written communication, interesting arguments, good shreds of evidence or supporting examples, structured organization, and good linguistic accuracy. The students can enhance those five sub-skills in the aspect of the ability to construct well-structured arguments due to the assistance of AI-assisted tools accompanying them in almost every writing step. For instance, the students used ChatGPT to brainstorm and develop arguments' reasons and examples. A similar study confirmed that the use of AI-assisted tools (e.g. ChatGPT or Perplexity) could offer interesting and complete information about the prompt given (Chen, 2023). Moreover, DeepL and Google Translate provided an easy and accurate language translation to help the students comprehend and display the information through writing (Kenny, 2019; Kureth et al., 2023). Quillbot might help them paraphrase their writing and correct the grammatical errors so their essays had good accuracy (Syahnaz & Fithriani, 2023). The existence of human intelligence provided controls in structuring the argument traits that made their argumentative essays more organized (Dhillon et al., 2024).

Second, this study also found that there was a significant improvement in the students' critical thinking skills. This study proved that the use of AI-assisted tools was not always a threat to decreasing students' critical thinking skills. This study evidenced that students were guided to practice their critical thinking skills through analyzing, synthesizing, interpreting, and evaluating relevant data obtained from AI to build appropriate reasons and evidence. Moreover, they could use different AI-assisted tools collaboratively and precisely. What we meant by a precise use reflected how students could critically and wisely use AI-assisted tools according to their needs. For example, they only used ChatGPT to obtain more information and check the linguistic accuracy. This critical awareness of different AI functions showed how students could benefit AI tools to support argumentative writing. Some studies also agreed that AI-assisted tools could be used by both teachers and students to train critical thinking skills (Liu & Wang, 2024; Malik et al., 2023; Namoun et al., 2024). However, limited studies still conveyed a strategy for permeating the teaching of critical thinking skills using AI-assisted tools. The present study provided insight into how to expose critical thinking skills within the teaching of argumentative writing. In this study, the ability to draw a non-fallacious conclusion to an argument was the most significantly improved aspect of critical thinking skills. Some studies believe that the making of inferences or conclusions had become the most difficult aspect of creating an argument due to confronting the ability to cooperate with claims, reasons, and evidence simultaneously (El Khoiri & Widiati, 2017; Guo, 2022). According to this study's result, the

use of AI-assisted tools might help students enhance their ability to draw a non-fallacious conclusion. Therefore, the success of mastering critical thinking skills could be assisted by the implementation of a human-AI collaborative strategy.

This study demonstrated that the human-AI collaborative writing strategy effectively fostered personalized learning, enhanced writing enjoyment, and improved writing efficacy among students. Personalized learning was operationally defined as students' ability to adapt both the selection and sequence of AI tools (ChatGPT, Google Translate, and Quillbot) to their individual writing needs during the cyclical process of argumentative essay development. Empirical evidence revealed that approximately 86.16% of students reported experiencing personalized learning, as they could customize their tool usage patterns - for instance, some frequently alternated between ChatGPT for idea generation and Quillbot for refinement, while others prioritized Google Translate for multilingual support before employing ChatGPT for structural improvements. This self-directed tool application, varying in both frequency and combination according to students' specific requirements at different writing stages (e.g. brainstorming, drafting, revising), constituted the core mechanism through which personalized learning was actualized in the study. The adaptive nature of this AI-assisted writing process thereby accommodated diverse student preferences and skill levels, confirming the strategy's effectiveness in promoting individualized writing development. Several prior studies regarding the existence of personalized learning confirmed this finding (Dhillon et al., 2024; Yeh et al., 2024). Another interesting finding was the fact that students experienced writing enjoyment and improved their writing efficacy. Some studies confirmed that AI-assisted tools might increase writing motivation due to the easiness and effectiveness offered (Fang et al., 2023; Guo & Li, 2024; Song & Song, 2023). This consequently increased their efficacy in writing due to the positive beliefs confessed that they could finish the writing of argumentative essays. Many studies, regardless the study fields, had agreed that the higher the motivation to finish the course tasks, the higher their efficacy (Anderson et al., 2023; Carmichael & Taylor, 2005; Song & Song, 2023).

Training students' critical thinking skills obviously required the role of teachers in conducting human-AI collaborative teaching. The existence of human intelligence as the focus of the collaborative teaching strategy gave the students experience in drilling their natural higher-order thinking skills (Nguyen et al., 2024). Teachers were suggested to be creative in utilizing AI-assisted tools to stimulate their students' scientific skills, as a part of enhancing their cognitive skills. For example, they could ask students to use AI tools to find a current issue, which was relevant to students' surrounding phenomena, that might need further

investigation. Additionally, they could arrange a certain teaching strategy to teach different English skills, such as listening, reading, and speaking. Of course, the involved AI-assisted tools could be various and material-context dependent to easily achieve the learning outcomes (Korteling et al., 2021). In an EFL speaking class, for instance, teachers can use ELSA Speak to practice pronunciation assisted by AI tools (Tran & Vu, 2024). Many studies also suggested that English teachers should be able to master devising human-AI collaboration-based teaching when delivering materials to avoid the drawbacks of AI reliance (Dai & Liu, 2024; Pratama et al., 2024; Wang, 2024).

## Conclusion

This study concludes that the human-AI collaborative writing strategy effectively improves Indonesian EFL students' argumentative writing skills. All subskills of the ability to construct a well-structured argument and convey critical thinking skills show a significant difference after the implementation of the human-AI collaborative writing strategy. Students also showcase their very good experience in writing argumentative essays using the undertaken strategy. They also state that the strategy promotes writing enjoyment and can heighten their writing efficacy. They also recommended the use of human-AI collaboration in any future learning process. Besides the strong improvements in students' argumentative writing skills and positive experiences with the human-AI collaborative writing strategy, this study admits several limitations that become the research gaps. First, the study only uses 30 participants, who become members of a debate community. With this participant number and characteristic, the study's conclusion could not be inclusive of diverse EFL student backgrounds. For example, it is necessary to prove whether the human-AI collaborative writing strategy worked well with the participants who have not been passionate about argumentative writing activities. Second, this study lacks a control group in the pre-experimental design, which prevents attribution of observed gains solely to the intervention. Third, the use of the same essay topic for pre- and post-testing introduces potential practice effects and content familiarity bias. Fourth, the short duration of the intervention and the absence of long-term follow-up offer limited information on how the strategy sustains its impact on students' argumentative writing and critical thinking skills. In this case, there is a need to examine the longitudinal impacts of the human-AI collaborative writing strategy. At last, this study does not reveal the students' writing motivation, which may be worth investigating as it affects the strong writing path to finish the argumentative writing. Future researchers can also focus on another writing genre, like narrative or story writing, to test the

effectiveness of implementing human-AI collaboration in the teaching and learning of critical thinking in the EFL context.

### Disclosure statement

No potential conflict of interest was reported by the author(s).

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## Appendix A1. Adapted Multiple Traits (MT) and Holistic Critical Thinking Scoring Rubric (HCTSR)

MT:

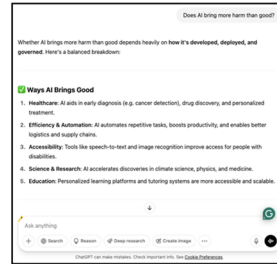
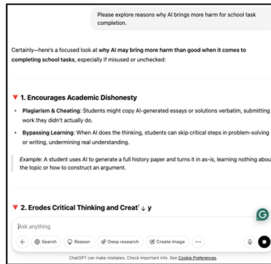
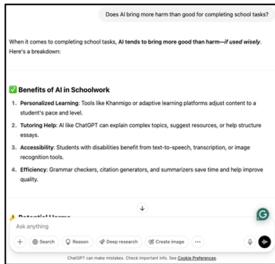
| Aspects               | Scores                                                                          |                                                              |                                                                                                      |                                                                                 |                                                                                                |                                                                                                     |                                                                                                      |                                                                                         |                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                       | 1                                                                               | 2                                                            | 3                                                                                                    | 4                                                                               | 5                                                                                              | 6                                                                                                   | 7                                                                                                    | 8                                                                                       | 9                                                                                                   |
| Communicative quality |                                                                                 | No ability to communicate                                    | The writing does not display an ability to communicate although meaning comes through spasmodically. | Limited ability to communicate, which puts a strain on the reader throughout    | The writing displays an ability to communicate although there is often strain for the readers. | The writing displays an ability to communicate although there is occasional strain for the readers. | The writing displays an ability to communicate with few difficulties for the reader.                 | The writing displays an ability to communicate without causing the reader difficulties. | The writing displays an ability to communicate in a way that gives the reader full of satisfaction. |
| Interestingness       |                                                                                 | Completely void of interesting content                       | Dull and uninteresting for most readers                                                              | Less interesting with limited new information                                   | The writing occasionally provides new information but little of it is interesting.             | The writing occasionally shows interesting ideas that attract reader attention.                     | The writing has frequent novel ideas that attract reader interest and attention.                     | The writing shows novelty and creativity, sustaining interest throughout.               | High creativity and novelty, fully engrossing the readers                                           |
| Referencing           | A true non writer who hasn't produced any assessable strings of English writing | No examples                                                  | No suitable examples                                                                                 | Fragmented examples that assist few readers                                     | The writing makes infrequent use of explanations of examples.                                  | The writing makes use of examples although they may not be culturally appropriate.                  | Many suitable examples for most readers                                                              | Frequent use of suitable examples for many readers                                      | Abundant use of illustrations and examples displaying cultural awareness.                           |
| Organization          |                                                                                 | No organizational structure or message is recognizable.      | No discernable organizational structure, and a message cannot be followed.                           | Lacks of clear organizational structure and the message is difficult to follow. | Well organized enough for the message to be followed most of the time                          | Well organized enough for the message to be followed throughout                                     | Good organizational structure for the message to be followed throughout                              | Good logical organizational structure that enables the message to be followed easily    | Completely logical organizational structure that enables the message to be followed effortlessly    |
| Linguistic accuracy   |                                                                                 | No control of vocabulary, spelling, punctuation, and grammar | Awareness of gross inadequacies of vocabulary, spelling, punctuation, and grammar                    | An existing control of vocabulary, spelling, punctuation, and grammar           | Awareness of errors of vocabulary, spelling, punctuation, and grammar that intrude frequently  | Awareness of errors of vocabulary, spelling, punctuation, and grammar—but only occasionally         | Awareness of but not troubled by occasional errors of vocabulary, spelling, punctuation, and grammar | No significant errors of vocabulary, spelling, punctuation, and grammar                 | No errors of vocabulary, spelling, punctuation, and grammar                                         |

HCTSR:

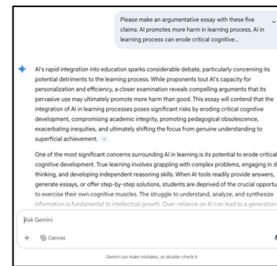
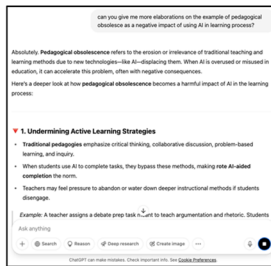
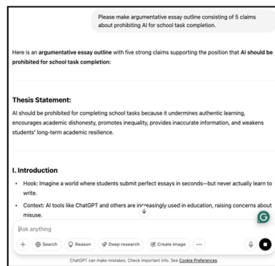
| Aspects                                                                                                                                                                                                                                                                   | Scores                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Accurately supporting the claim with relevant reasons, evidences, and inferences<br>Avoiding logical fallacies<br>Providing and explaining alternatives and their possible impacts<br>The coherence of the logic between paragraphs<br>Drawing non-fallacious conclusions | Consistently does all or almost all of the following:<br>Offers biased interpretations of evidence, statements, graphics, questions, information, or the points of view of others. Fails to identify or hastily dismisses strong, relevant counter-arguments. Ignores or superficially evaluates obvious alternative points of view. Argues using fallacious or irrelevant reasons, and unwarranted claims. Does not justify results or procedures, nor explain reasons. Regardless of the evidence or reasons, maintains or defends views based on self-interest or preconceptions. Exhibits closed-mindedness or hostility to reason.<br><b>(Significantly weak critical thinking skills)</b> | Does most or many of the following:<br>Misinterprets evidence, statements, graphics, questions, etc. Fails to identify strong, relevant counter-arguments. Ignores or superficially evaluates obvious alternative points of view. Draws unwarranted or fallacious conclusions. Justifies few results or procedures, seldom explains reasons. Regardless of the evidence or reasons, maintains or defends views based on self-interest or preconceptions. <b>(Unacceptable critical thinking skills)</b> | Does most of many of the following:<br>Accurately interprets evidence, statements, graphics, questions, etc. Identifies relevant arguments (reasons and claims) pro and con. Offers analyses and evaluations of obvious alternative points of view. Draws warranted, non-fallacious conclusions. Justifies some results or procedures, explains reasons. Fair-mindedly follows where evidence and reasons lead. <b>(Acceptable critical thinking skills)</b> | Consistently does all or almost all of the following:<br>Accurately interprets evidence, statements, graphics, questions, etc. Identifies the most important arguments (reasons and claims) pro and con. Thoughtfully analyzes and evaluates major alternative points of view. Draws warranted, judicious, non-fallacious conclusions. Justifies key results and procedures, explains assumptions and reasons. Fair-mindedly follows where evidence and reasons lead. <b>(Strong critical thinking skills)</b> |

## Appendix A2. Examples of Students' Writing Process with Human-AI Collaborative Writing Strategy

### Brainstorming Stage



### Outlining and Developing Stages



### Revision and Editing Stages

